



Fragments of Growth User Manual

Course: CS2206 - Computer Graphics 2026
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1. Project Overview

Fragments of Growth is an interactive narrative game built using C++ and OpenGL. The story follows a character through a symbolic world that starts in grayscale. As the player collects glowing fragments and solves puzzles across various environments, the world regains its color, representing the journey of self-discovery and acceptance.

2. Setup and Installation

To build and run the project, ensure your environment meets the following requirements:

- **Compiler:** C++ compiler (Visual Studio 2022 recommended).
- **Libraries:**
 - **GLEW:** For managing OpenGL extensions.
 - **FreeGLUT:** For window handling and user input.
- **Asset Paths:** Ensure the bitmap textures (cover.bmp and end.bmp) are placed in the paths defined in the code, or update the image1Path and image2Path variables to match your local directory.

3. Controls & Interaction

Action	Key / Input
Start Game	ENTER (at Cover Screen)
Move Character	Arrow Keys (Up, Down, Left, Right)
Next Scene	N (once the current puzzle is solved)
Reset Scene	R (restarts the current level)
About/Help	A for About / H for Help
Special Interaction	V, X, Z, or U (used during the Desert scene)
Library Puzzle	Mouse Click & Drag (to arrange books)
Exit	ESC



4. Scene Objectives

The project is divided into multiple scenes, each representing a different stage of growth:

1. **Forest Path (Identity):** Navigate the woods and find the real glowing fragment to restore color to nature.
2. **Desert (Courage):** Face a storm and choose the path of bravery through specific keyboard choices.
3. **The Beach (Clarity):** Reach the hidden "Clarity" zone to unlock the fragment.
4. **The City (Connection):** Walk near nodes to activate lights and reconnect the urban landscape.
5. **The Library (Purpose):** Use the mouse to drag and drop books into alphabetical order (A through F) to find purpose.
6. **The Garden (Acceptance):** Collect the final set of fragments to conclude the story.